

Name: LLOYD S. HYATTE

Course & Year: BSIT-3f

Subject: Web Development / API Integration

1. What is a REST API?

A REST API is an application programming interface that conforms to the design principles of the REST architectural style. It acts as a bridge that allows two different software systems to communicate with each other over the internet. When we build a web application, the REST API is what enables the frontend interface to request and send data to the backend database securely and efficiently.

2. What does REST stand for?

REST stands for **Representational State Transfer**.

3. What is the difference between a client and a server?

The **client** is the application or device that the user interacts with—such as a web browser loading a website, or a mobile app interface. It sends requests for data or actions. The **server** is the remote computer or system that listens for those requests, processes the underlying logic, interacts with the database (like SQL), and sends the appropriate response back to the client.

4. List and explain at least four HTTP methods.

HTTP methods define the action the client wants the server to perform. The four main methods (often mapping to CRUD operations) are:

- **GET:** Used to request or retrieve data from a specific resource. It should only fetch data and not change anything on the server.
- **POST:** Used to send new data to the server, typically to create a brand new resource (e.g., submitting a form to register a new user).
- **PUT:** Used to update an existing resource entirely. It replaces the current data at the target endpoint with the payload sent in the request.

- **DELETE:** Used to remove or delete a specific resource from the server.

5. What is a resource in a REST API?

A resource is any discrete piece of information or object that the API can manage. In a system like a canteen management app, resources could be "MenuItems", "Orders", or "Users". Each resource is identified by a unique URL and manipulated using the standard HTTP methods.

6. What is the purpose of a URL/endpoint?

An endpoint is the specific digital location or web address (URL) where an API receives requests about a specific resource. Its purpose is to direct the client's HTTP request to the exact function on the server that handles that particular data. For example, the endpoint `/api/users` directs the system to the logic managing user accounts.

7. What is JSON and why is it commonly used in REST APIs?

JSON stands for **JavaScript Object Notation**. It is a lightweight data-interchange format formatted as plain text. It is heavily used in REST APIs because it is very easy for humans to read and write, and equally easy for machines to parse and generate. Since it represents data as key-value pairs, it integrates seamlessly when connecting a backend (like PHP) to a frontend built with HTML, CSS, and JavaScript.

8. What does "stateless" mean in REST architecture?

In REST architecture, "stateless" means that the server does not store any information or memory about previous requests from the client. Every single HTTP request made by the client must contain all the information, authentication, and context the server needs to understand and process that specific request independently.